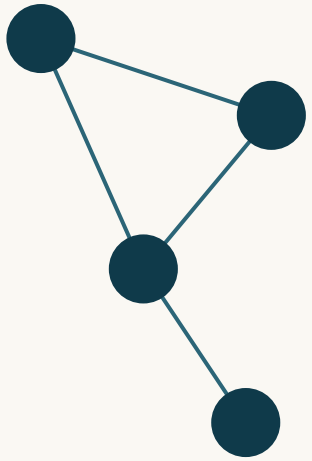
QUANTUM OPTIMIZATION

QAOA for MaxCut



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Physics 565 / 656 · Spring 2026

CHAPTER

01

Introduction

Graphs, cuts, and the MaxCut problem

Definition of a graph

A finite vertex set plus a set of unordered pairs

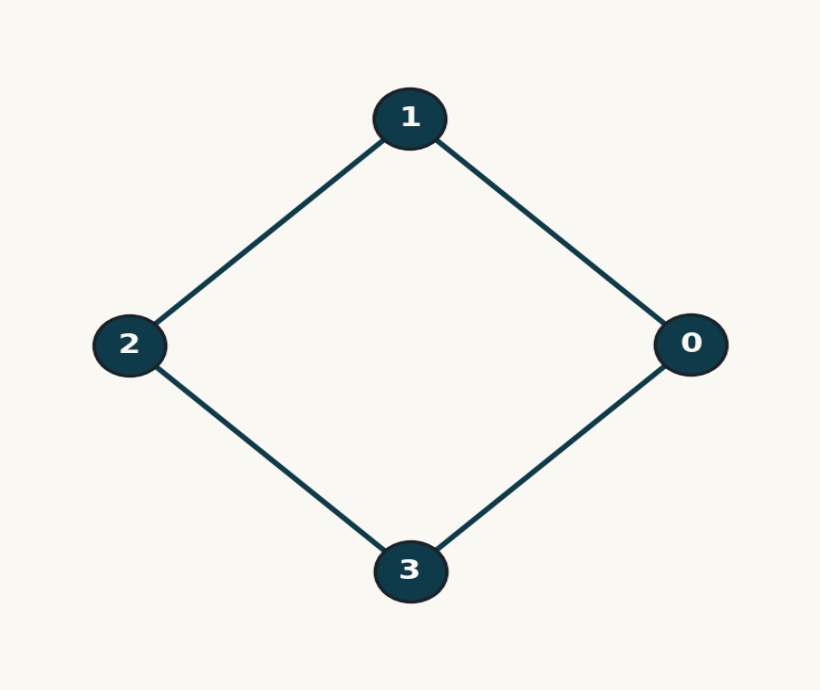
Vertex set

$V = \{0, 1, 2, 3\}, \qquad |V| = n = 4$

Edge set

$E \subseteq (V \textit{ choose } 2)$
$= \{ \{0,1\}, \{1,2\}, \{2,3\}, \{0,3\} \}$

C_4 : cycle on 4 vertices

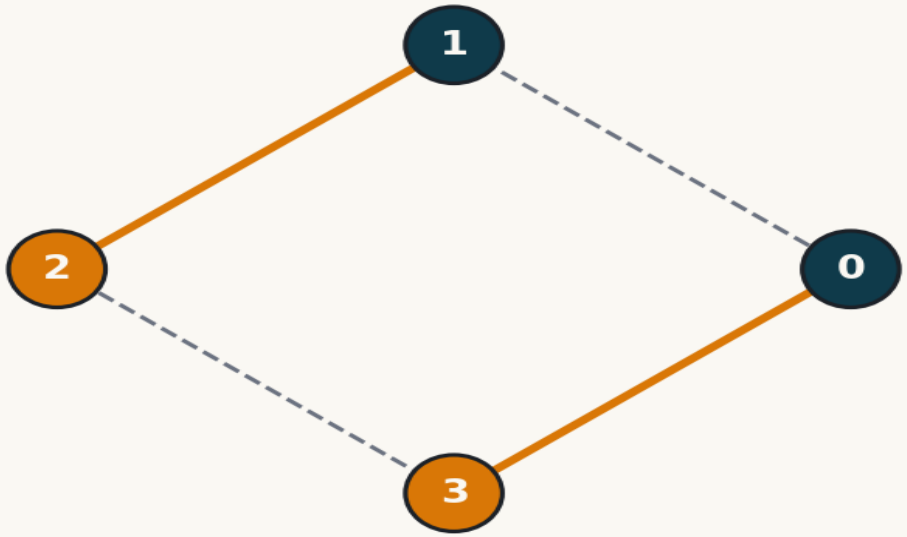


$n = 4, \quad |E| = 4$

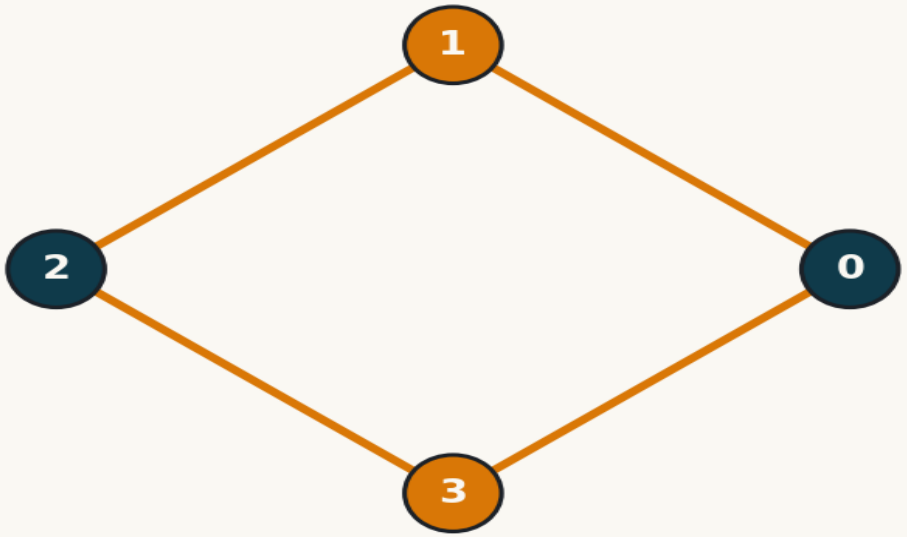
The MaxCut problem

Partition $V = S \sqcup S$ to maximise the number of cut edges

Suboptimal: $S = \{0, 1\}$ cut = 2



MaxCut: $S = \{0, 2\}$ cut = 4



***For C_4: 2^4** = 16 labelled partitions / 2^3 = 8 distinct cuts. The maximum is **4**.*

CHAPTER

02

Motivation

Why MaxCut matters beyond the textbook

Why MaxCut? Portfolio diversification

Splitting a portfolio so that each half is internally uncorrelated

1 The setup

n assets with daily returns $r_i(t)$. Pairwise correlation

$\rho_{ij} = Cov(r_i, r_j) / (\sigma_i \sigma_j) \in [-1, 1]$

2 Investor goal

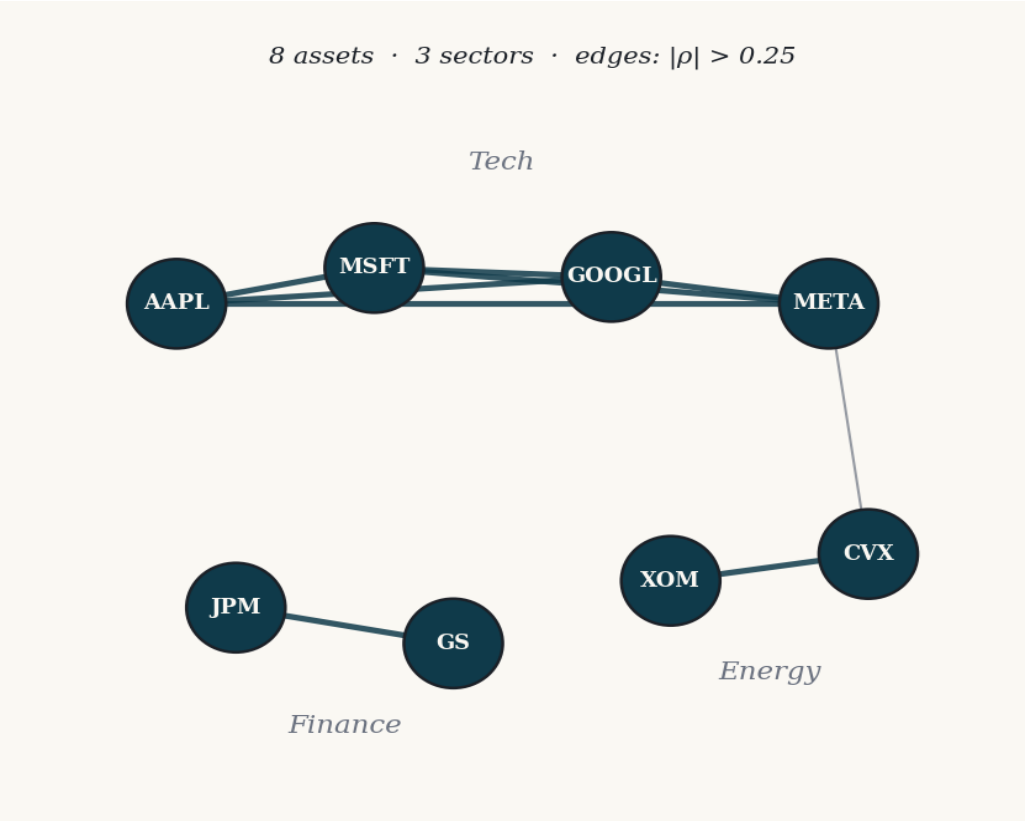
Partition assets into two risk groups S and S with

low correlation inside each group.

3 Why this is MaxCut

Build graph: vertex = asset, edge = $|\rho_{ij}| > 0.25$.

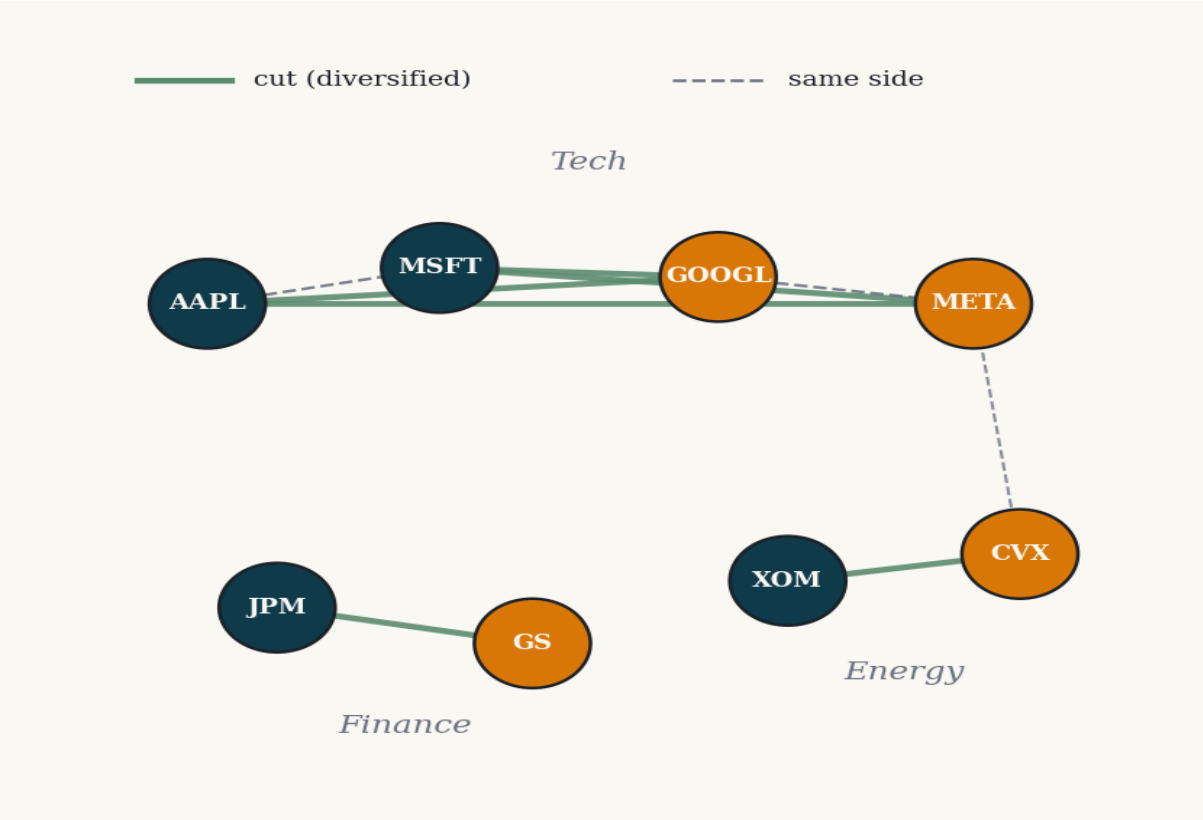
Cut edge ⇒ correlated pair is separated. max C(S) separates as many pairs as possible.



Intra-sector edges are thick (ρ ≈ 0.87). Finding the MaxCut separates those correlated pairs.

MaxCut gives the diversified partition

Correlation inside each group drops sharply — both halves become well-diversified



What the investor gets

Group S: AAPL, MSFT, JPM, XOM
Group S: GOOGL, META, GS, CVX

Each group contains one asset from every sector.

Strategy	Avg |ρ| within a group
Naïve: Tech | Fin+Energy	0.87 / 0.32
MaxCut	**0.22 / 0.26**

~4× less correlated inside each group — true diversification in both halves.

Same framework applies to:

- Job scheduling
- Facility location
- Network partitioning
- Circuit layout

CHAPTER

03

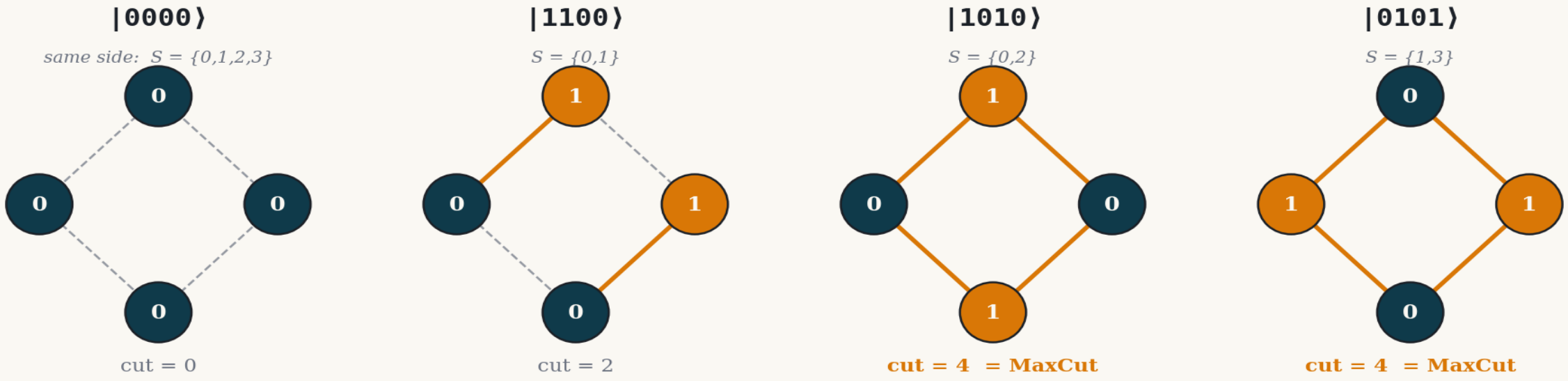
Theory

From the cost function to a cost Hamiltonian

What are we looking for?

A bitstring that encodes a MaxCut partition — shown on C_4

Measurement on $|\psi\rangle$ gives a bitstring $z = z_0 z_1 \cdots z_{n-1}$. Rule: $z_i = 0 \Rightarrow i \in S$, $z_i = 1 \Rightarrow i \in S$



Goal: *maximise the probability of sampling* $|1010\rangle$ *or* $|0101\rangle$ *(the two MaxCut solutions of* C_4*).*

Cost Hamiltonian: from cut to operator

Three steps: binary decision variable → spin variable → Pauli-Z operator

1

Decision variable

$z_i \in \{0, 1\}, \quad z_i = 0 \iff i \in S, \quad z_i = 1 \iff i \in S$

$$C(z) = \Sigma\, \mathbb{1}[z_i \neq z_j]$$

2

Rewrite with spin variable

$s_i = 1 - 2z_i \in \{+1, -1\}$

$$\mathbb{1}[z_i \neq z_j] = (1 - s_i\, s_j) / 2$$

3

Quantise

$s_i \rightarrow Z_i$ *(Pauli-Z on qubit i)*

$$H_C = \Sigma\ (I - Z_i\, Z_j) / 2$$

Starting state: $|+\rangle$ on all qubits

Four reasons this is the natural initial state for QAOA

$$|+\rangle^{\otimes n} = (1/\sqrt{2^n}) \sum_{z \in \{0,1\}^n} |z\rangle = H^{\otimes n} |0\rangle^{\otimes n}$$

1 Uniform over candidates

Equal weight on all 2^n bitstrings — no partition is preferred at the s tart.

2 Aligned with the mixer

$X|+\rangle = +|+\rangle$, so $H_B|+\rangle^{\otimes n} = n|+\rangle^{\otimes n}$ (eigenstate).

3 Cheap to prepare

One Hadamard per qubit — constant-depth circuit from $|0\rangle^{\otimes n}$.

4 Adiabatic starting point

Ground state of $H(t=0) = H_B$ in the adiabatic interpolation.

Classical benchmarks: greedy and GW

The baselines QAOA must compete with — GW is the polynomial-time gold standard

Greedy

Process vertices in order; each joins the side that cuts more already-assigned e dges.

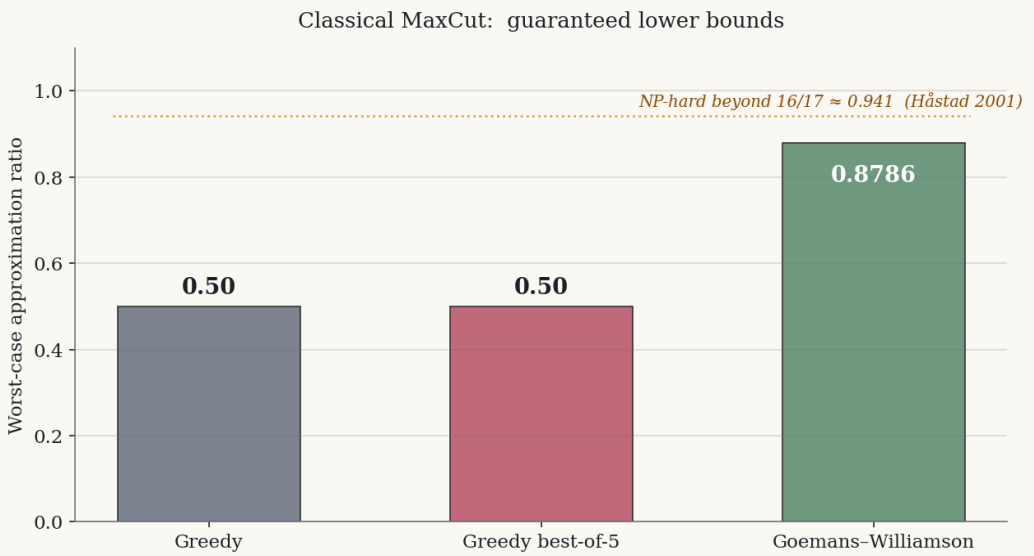
Greedy (best of 5)

Run greedy five times with different orderings; keep the best cut found.

Goemans–Williamson

Relax to unit vectors, solve SDP, round by random hyperplane. Runtime $O(n^{3.5})$.

Approximation ratio (lower bound)



CHAPTER

04

Implementation

Quantum circuit · Optimization · Warm start

Quantum circuit on C_{10}

Alternating layers of the problem and mixer unitaries

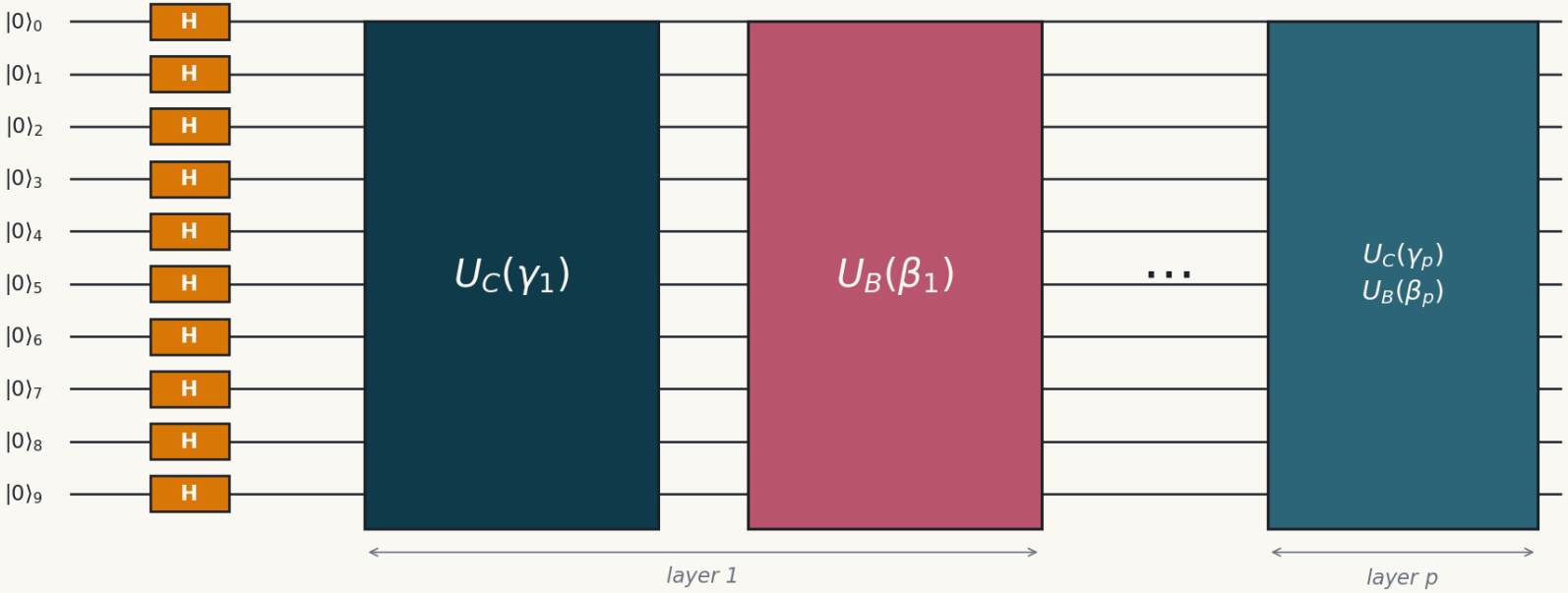


Figure. The QAOA ansatz applies p alternating blocks after the uniform superposition.

Ansatz state

$$|\psi_p(\gamma,\beta)\rangle$$

$$=U_B(\beta_p)U_C(\gamma_p)\cdots U_B(\beta_1)U_C(\gamma_1)|+\rangle^{\otimes 10}$$

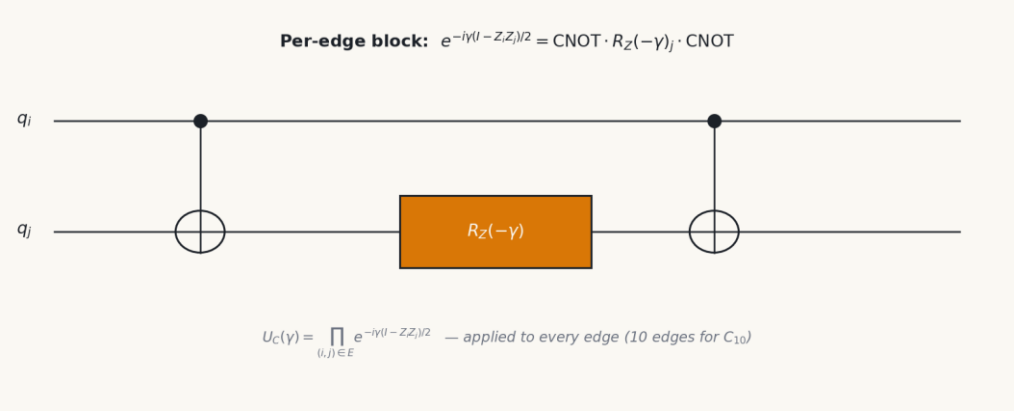
- 10 qubits (one per vertex)
- 2p variational parameters
- Initial state: uniform $|+\rangle$ on all qubits

Layer unitaries U_C and U_B

Problem Hamiltonian evolution and transverse-field mixer

U_C(γ) — Problem unitary

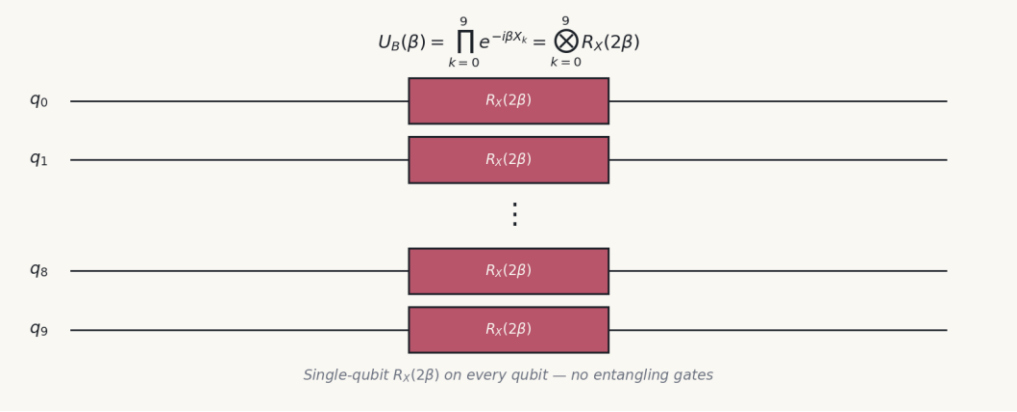
$U_C(\gamma) = e^{-i\gamma H_C} = \prod_{(i,j)\in E} e^{-i\gamma(I - Z_i\, Z_j)/2}$



- 10 CNOT·R_Z(−γ)·CNOT blocks per layer
- 20 CX + 10 R_Z gates per U_C

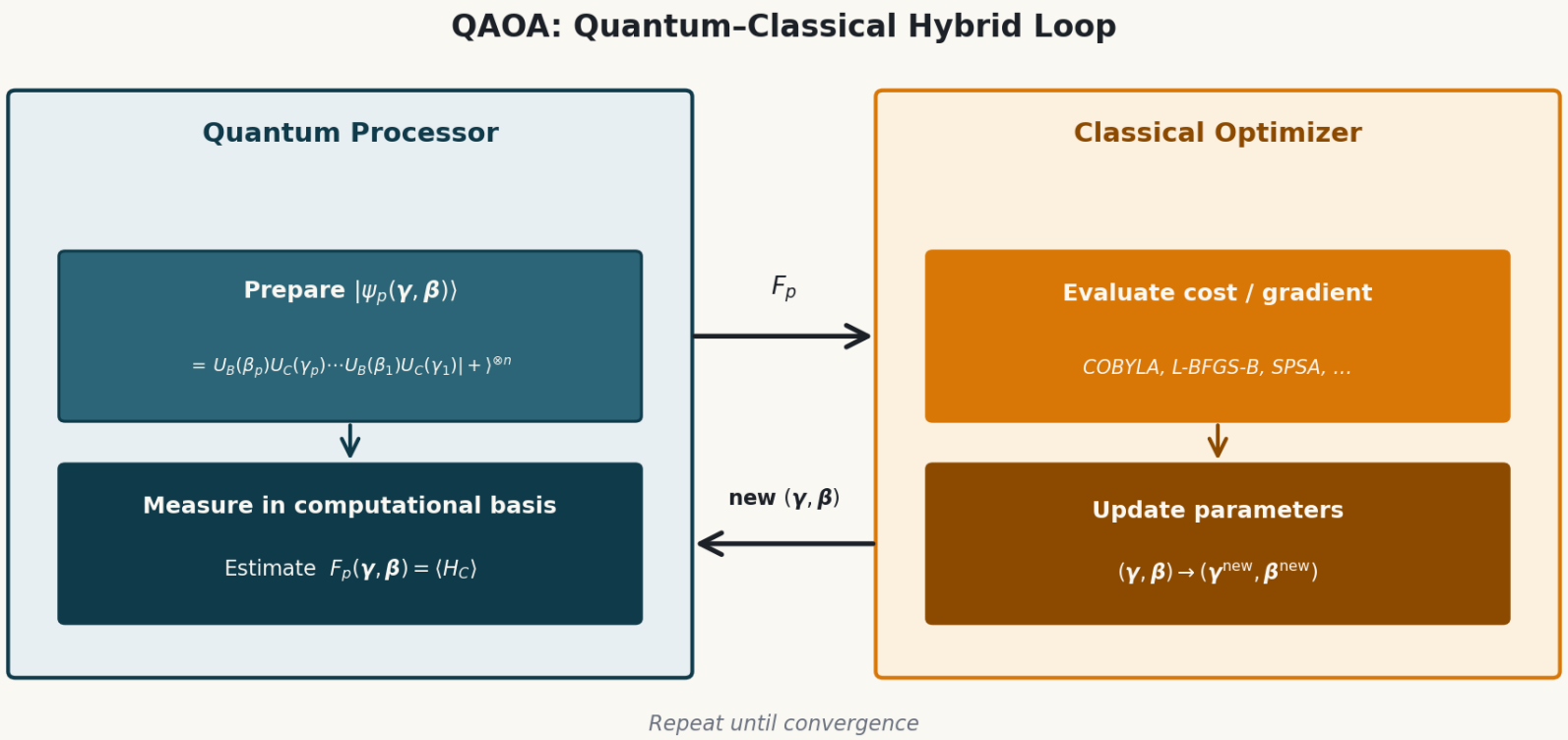
U_B(β) — Mixer unitary

$U_B(\beta) = e^{-i\beta H_B}, \quad H_B = \Sigma_k X_k$



- 10 single-qubit R_X(2β) gates per layer
- No entangling gates in U_B

Optimization as a hybrid loop



Each iteration

1. **Prepare** $|\psi_p(\gamma,\beta)\rangle$
2. **Measure** $\langle H_C \rangle$ via sampling
3. **Classical optimizer** returns new (γ, β)
4. **Repeat until** $|\Delta F_p| < \varepsilon$

Only cost values flow to the classical side — g radients are built from parameter-shift evalu ations.

Choosing the classical optimizer

C_{10} + 3 chords, $p = 3$ — statevector simulation

Optimizer	Gradient	Cost/step	Best for
COBYLA	None (trust region)	O(p)	Shot-noise robust
L-BFGS-B	Per-edge parameter-shift	2(|E|+n)p	Statevector, exact gradient
Nelder–Mead	None (simplex)	O(p)	Fallback when COBYLA stalls
SPSA	Stochastic 2-evaluation	2	Real hardware

***L-BFGS-B** converges fastest on noiseless statevector via the exact per-edge parameter-shift gradient. **COBYLA** i s run in parallel as a gradient-free cross-check, and would be the primary choice under shot noise. **SPSA** would replace both on real hardware.*

Our choice

L-BFGS-B

Exact gradient, fewest evals

+

COBYLA

Gradient-free baseline

Both with warm-start initialisation

Cross-check: two independent optimizers agr eeing on the same minimum rules out optimi zer-specific artefacts.

Warm start

Initialise p+1 layers from the p-layer optimum

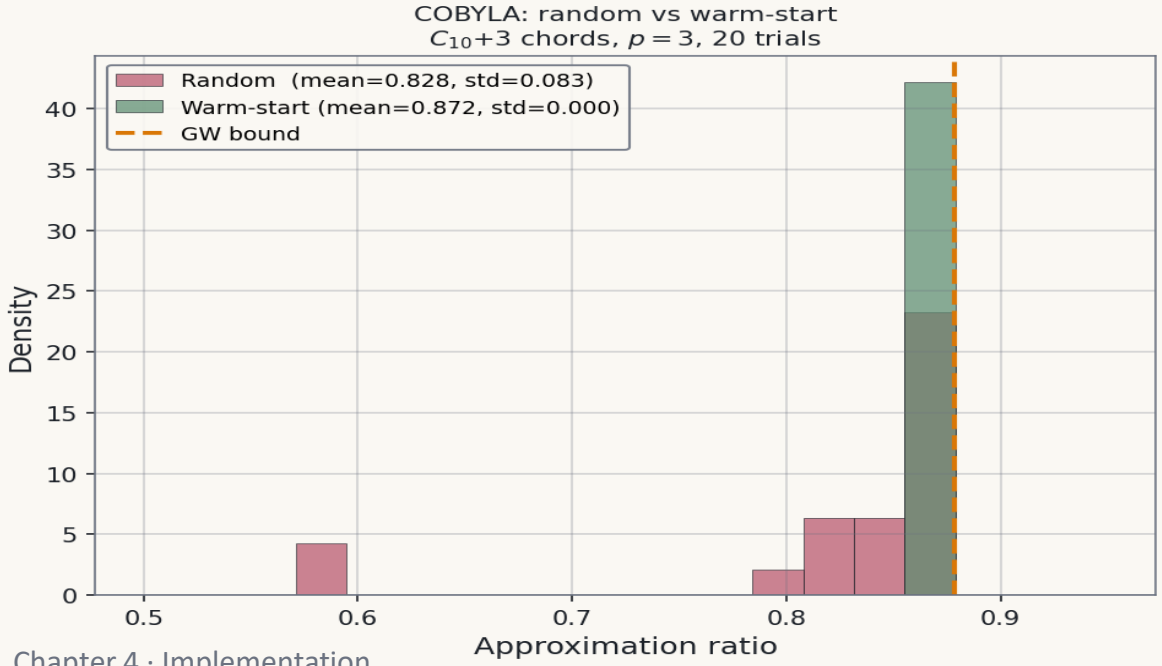
Intuition

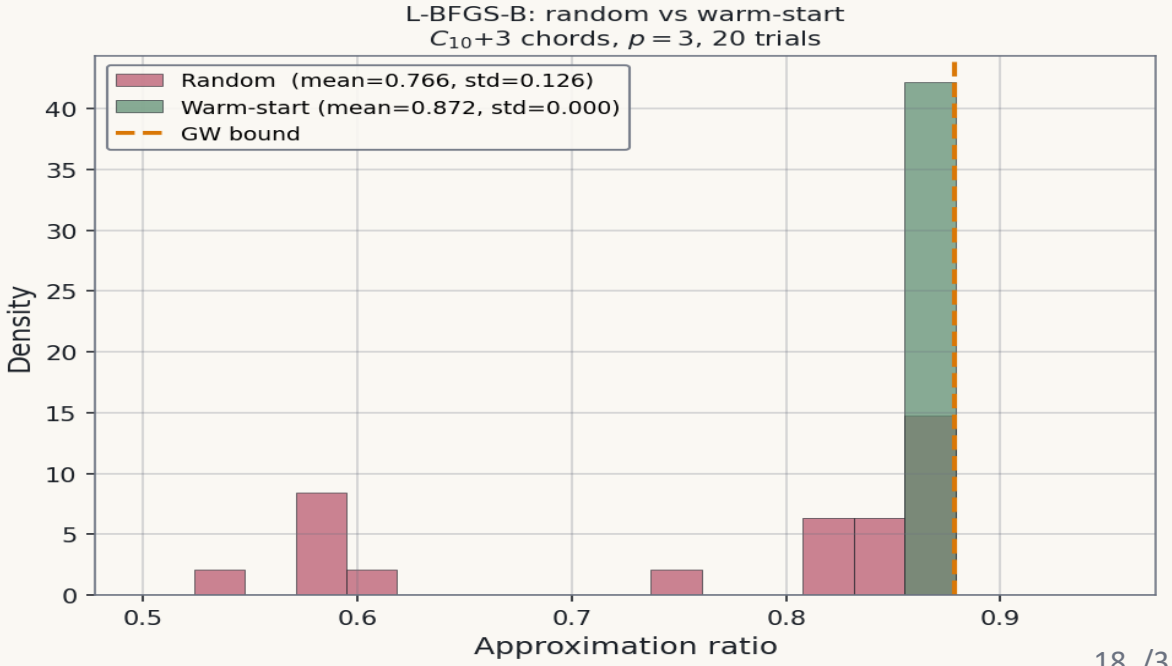
A p-layer optimum is already a good region of parameter space for p+1 layers — adding one layer with small γ, β is nearly the identity.

Random restarts at p=3 waste evaluations on distant local maxima.

Algorithm (Zhou et al. 2020)

```
1.  Optimise p = 1 globally (random restarts)
2.  For k = 2, …, p:
        θ₀ ← (γ*, γ*[−1]) ‖ (β*, β*[−1]) + η
        Local optimise F_k from θ₀
3.  Return final (γ*, β*)
```





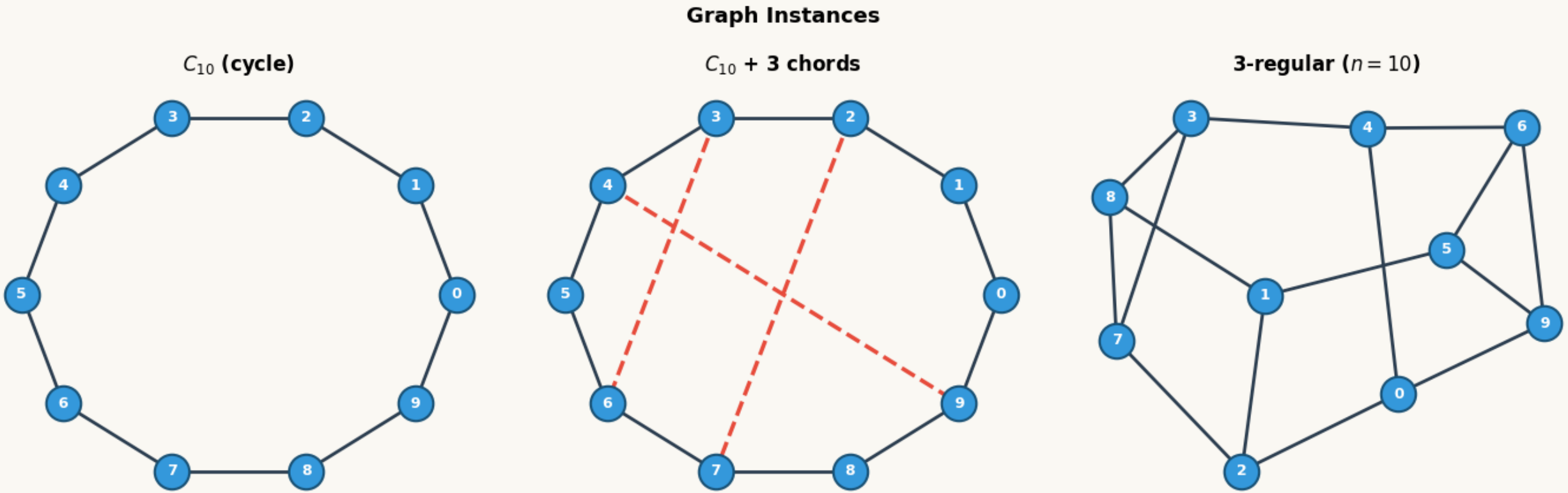
CHAPTER

05

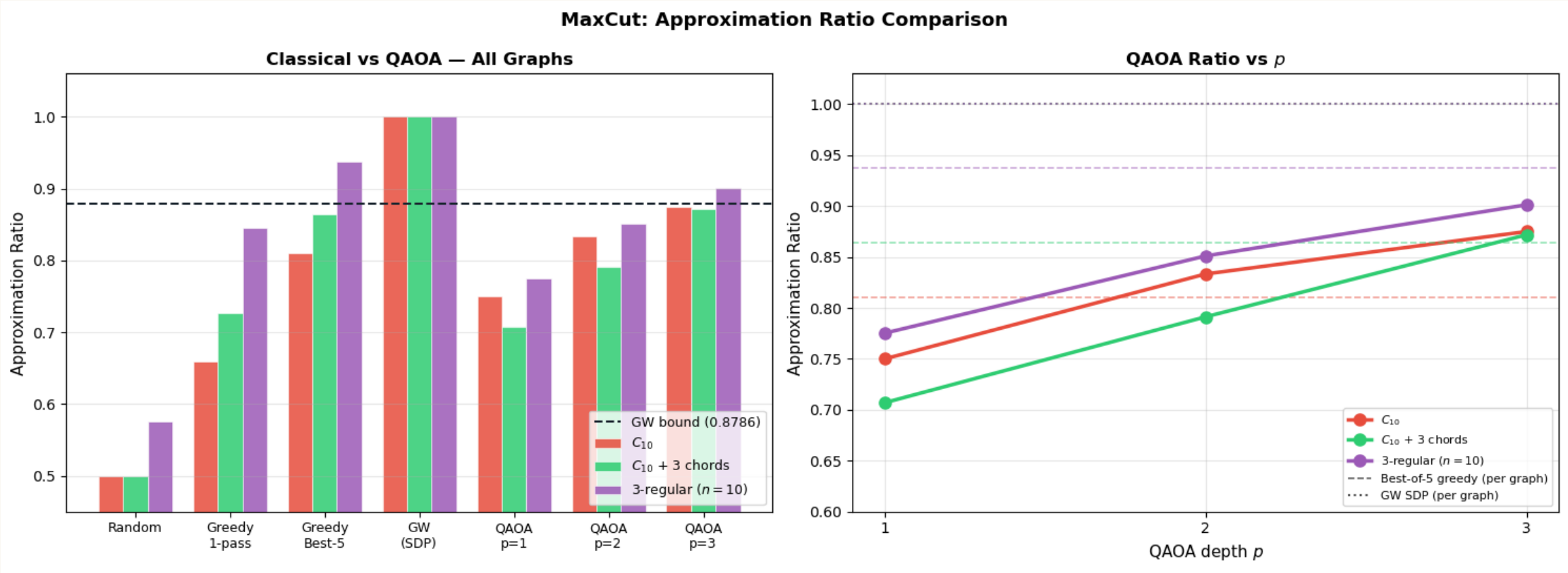
Results

Comparison Performance · Quantum Circuit · Measurements · Noise

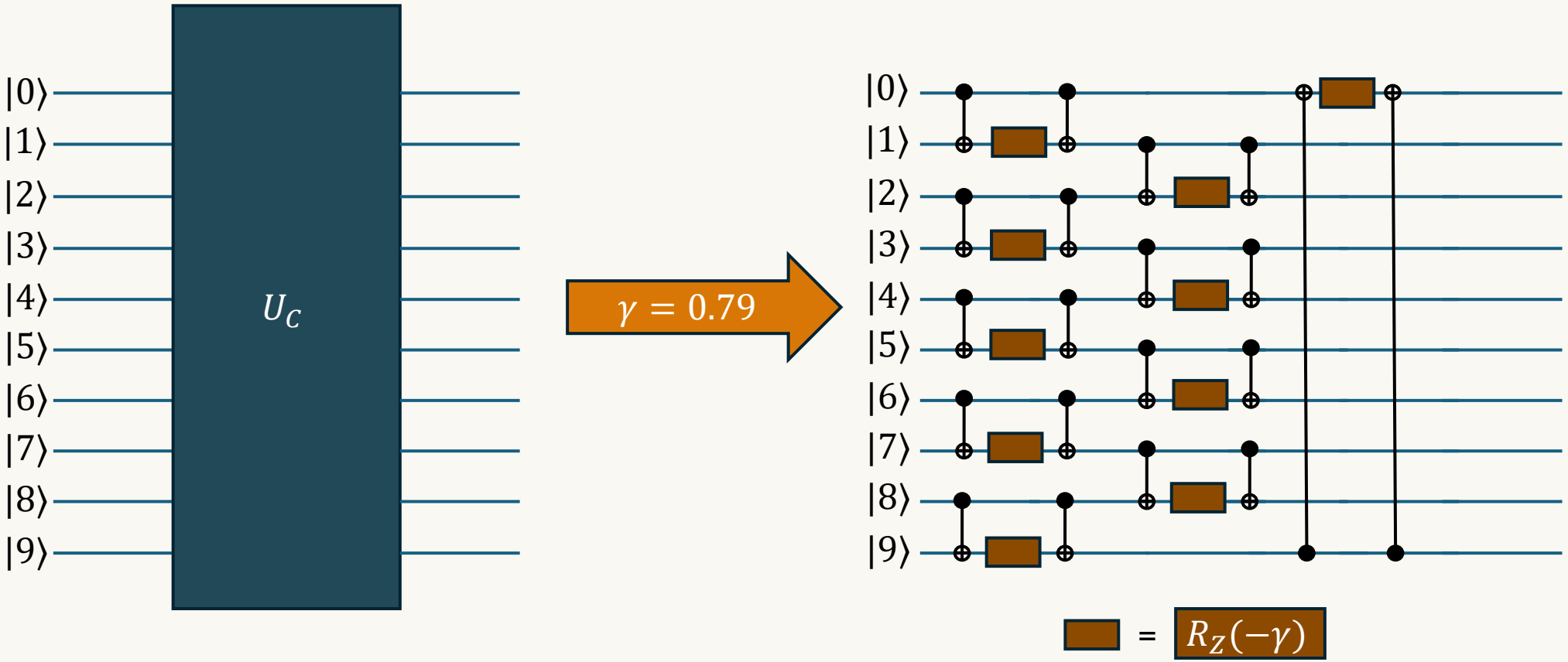
Problems to solve



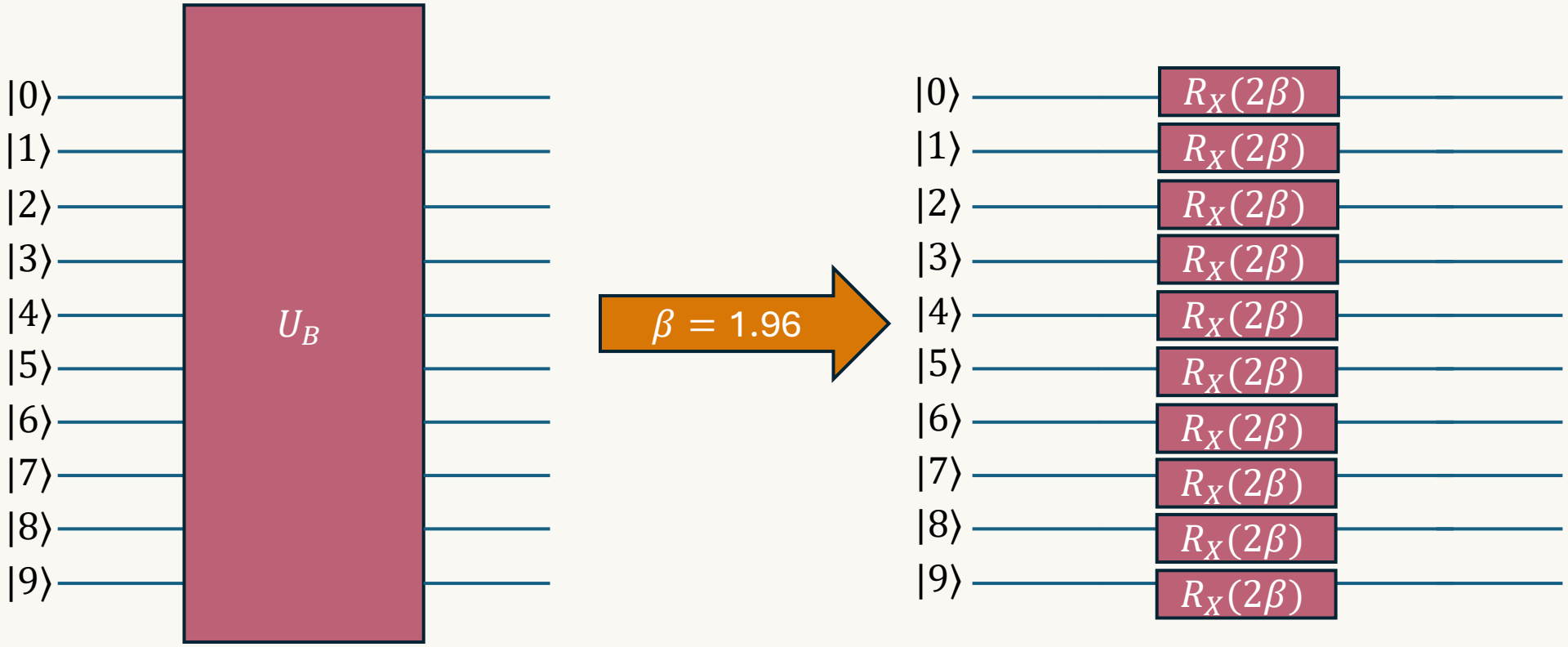
Performance Comparison



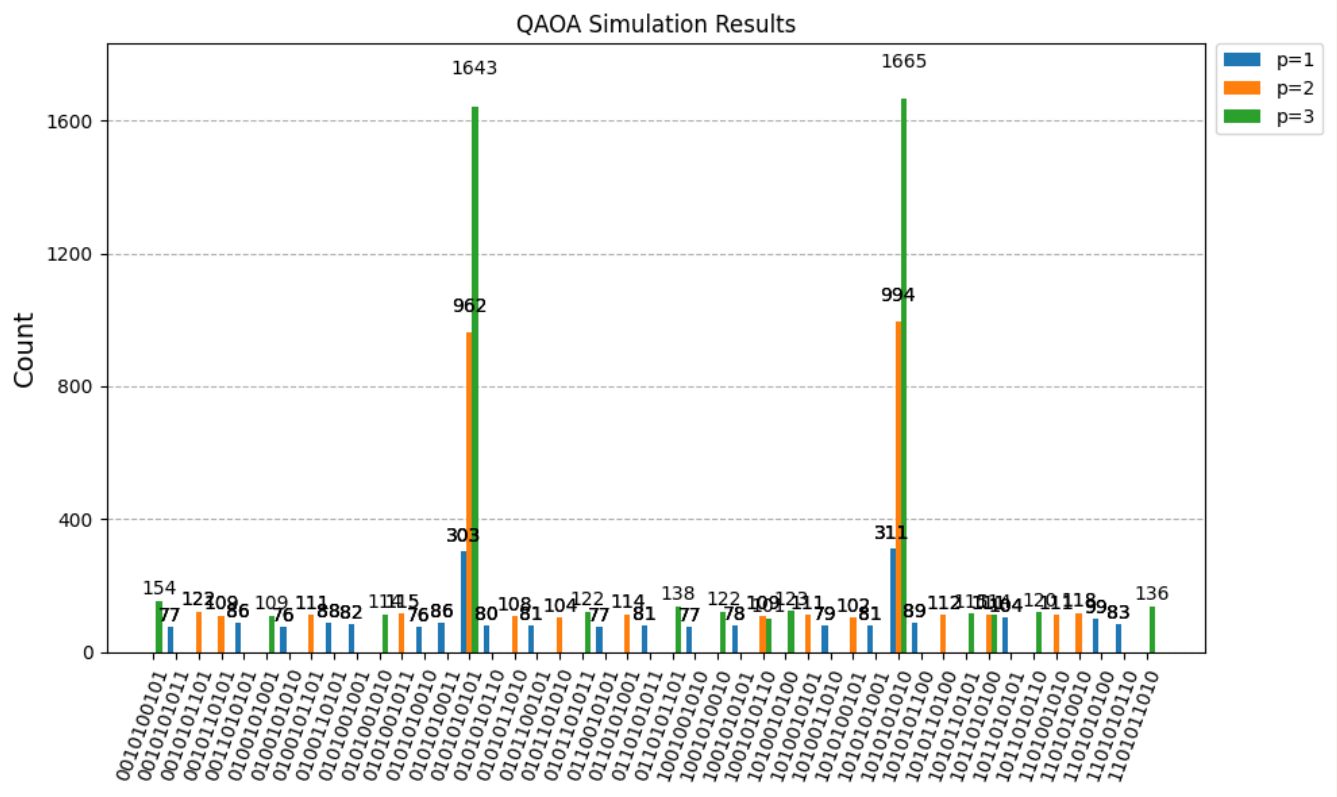
C_{10} Quantum Circuit – U_C Gate



C_{10} Quantum Circuit – U_B Gate

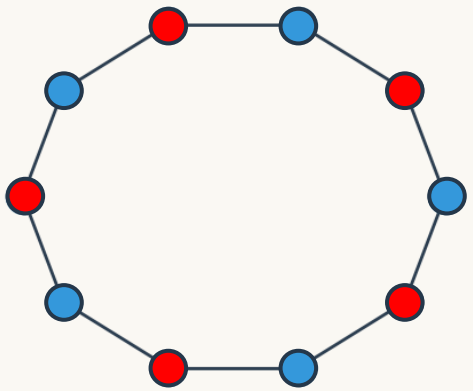


C_{10} Quantum Circuit – Measurement

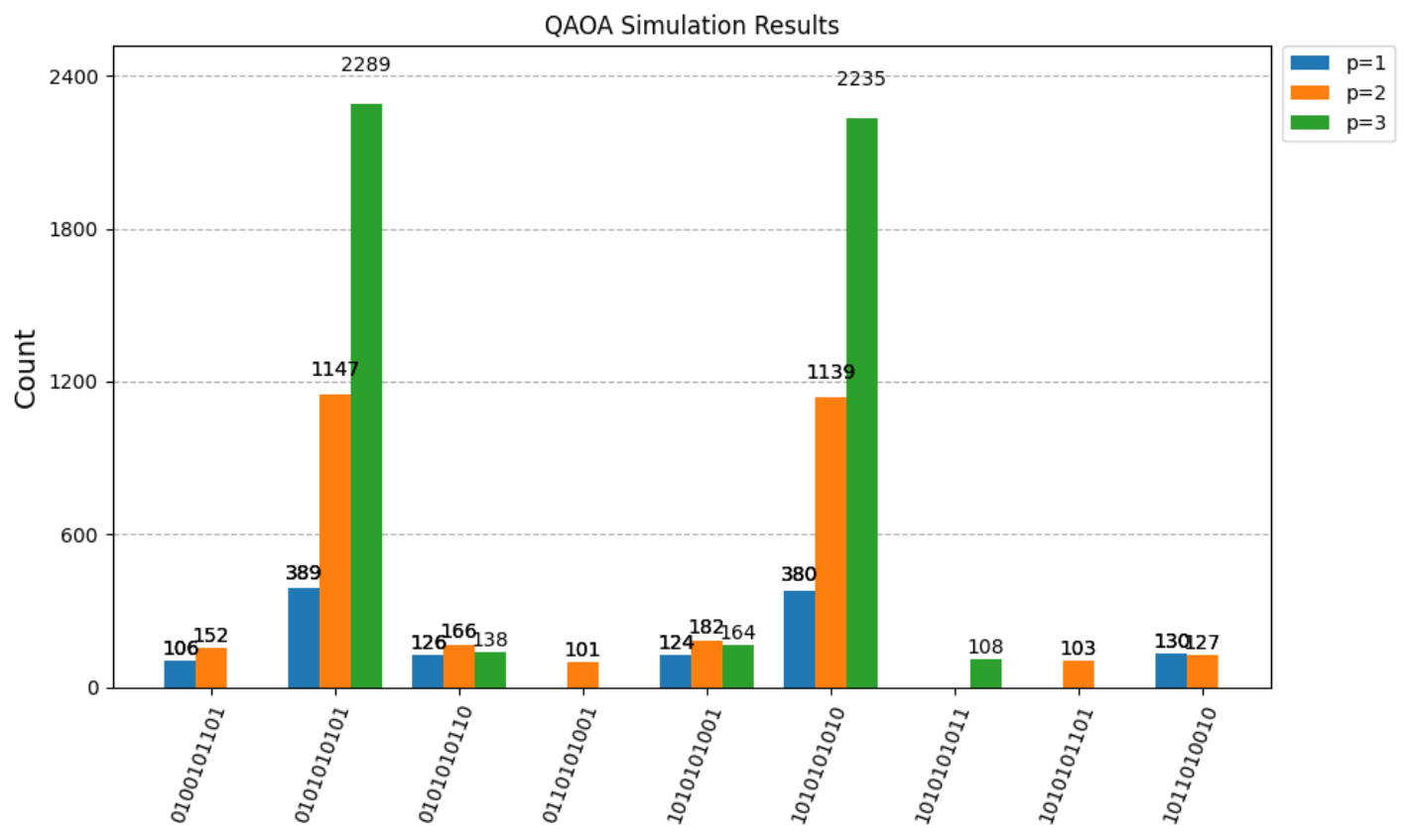


Measured states

- 1010101010
- 0101010101

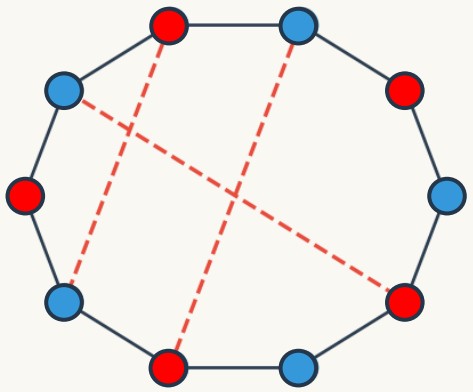


C_{10} +3 Quantum Circuit – Measurement

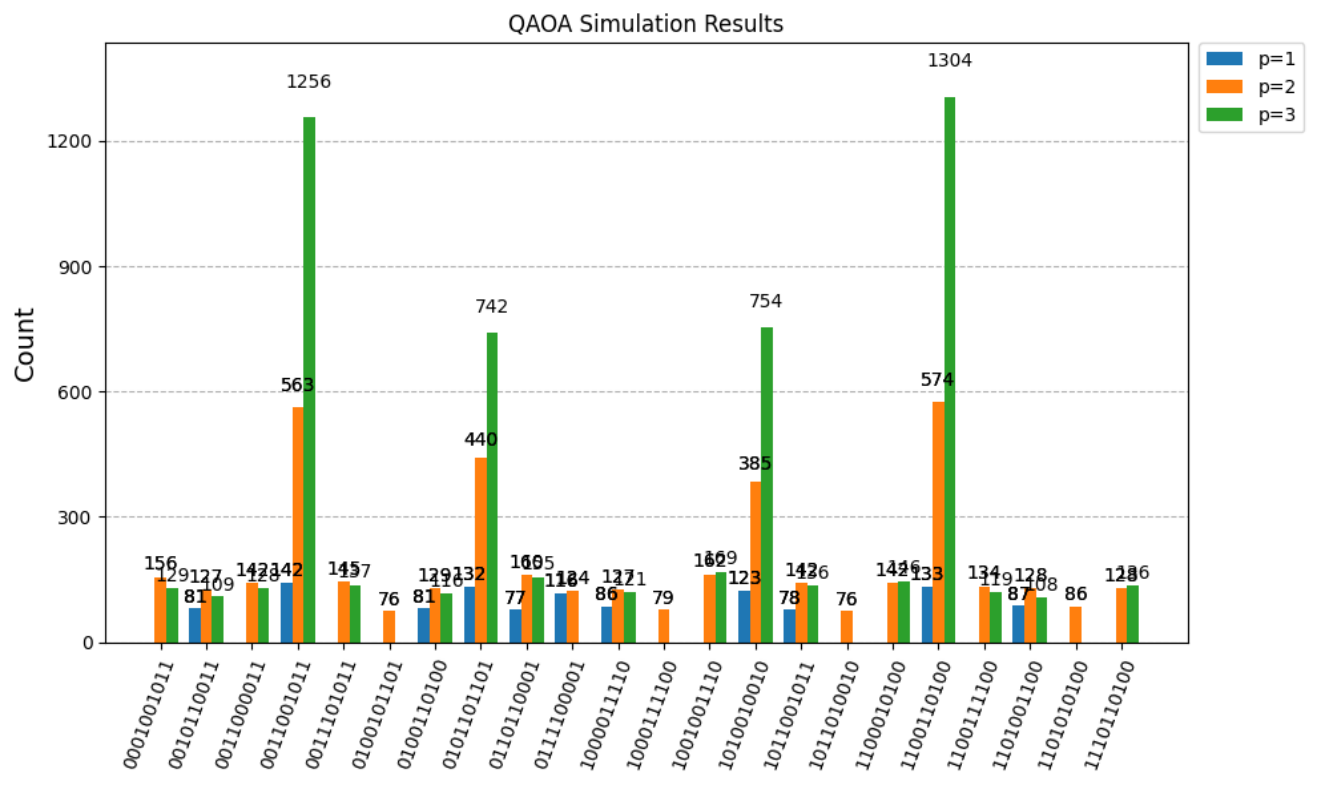


Measured states

- 1010101010
- 0101010101

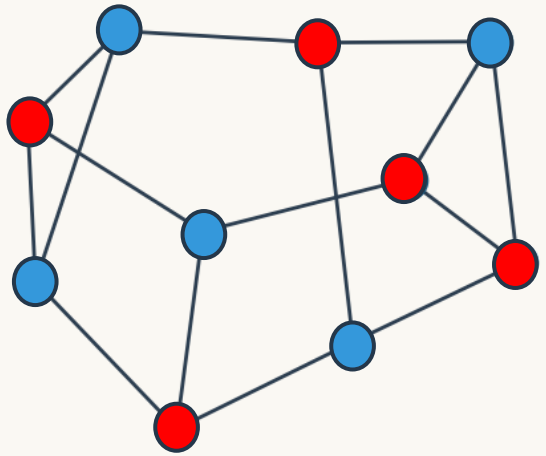


3-regular Quantum Circuit – Measurement

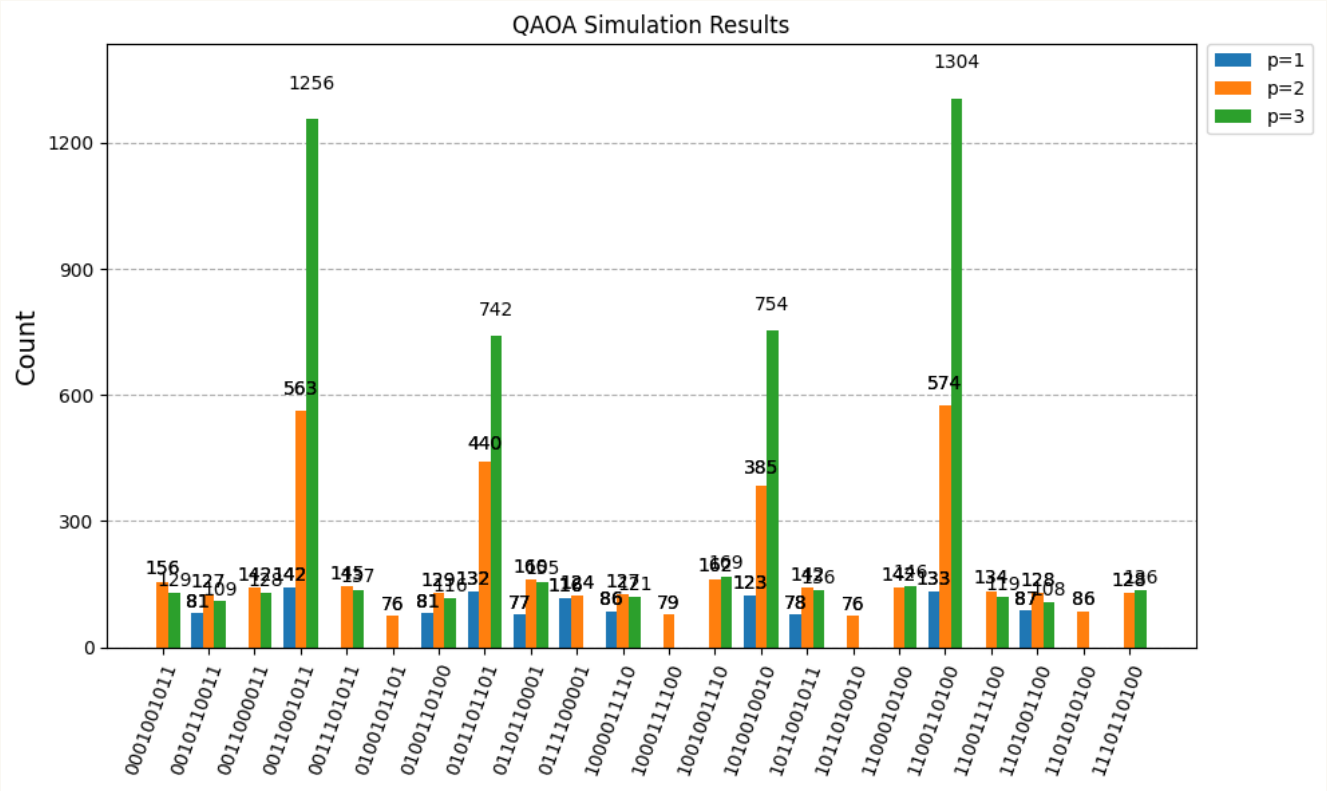


Measured states

- *MaxCut = 13*
- 1100110100
- 0011001011

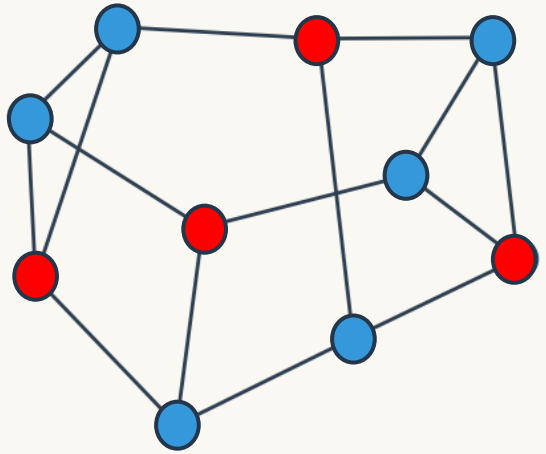


3-regular Quantum Circuit – 2nd Best



Measured states

- *MaxCut = 12*
- 0101101101
- 1010010010



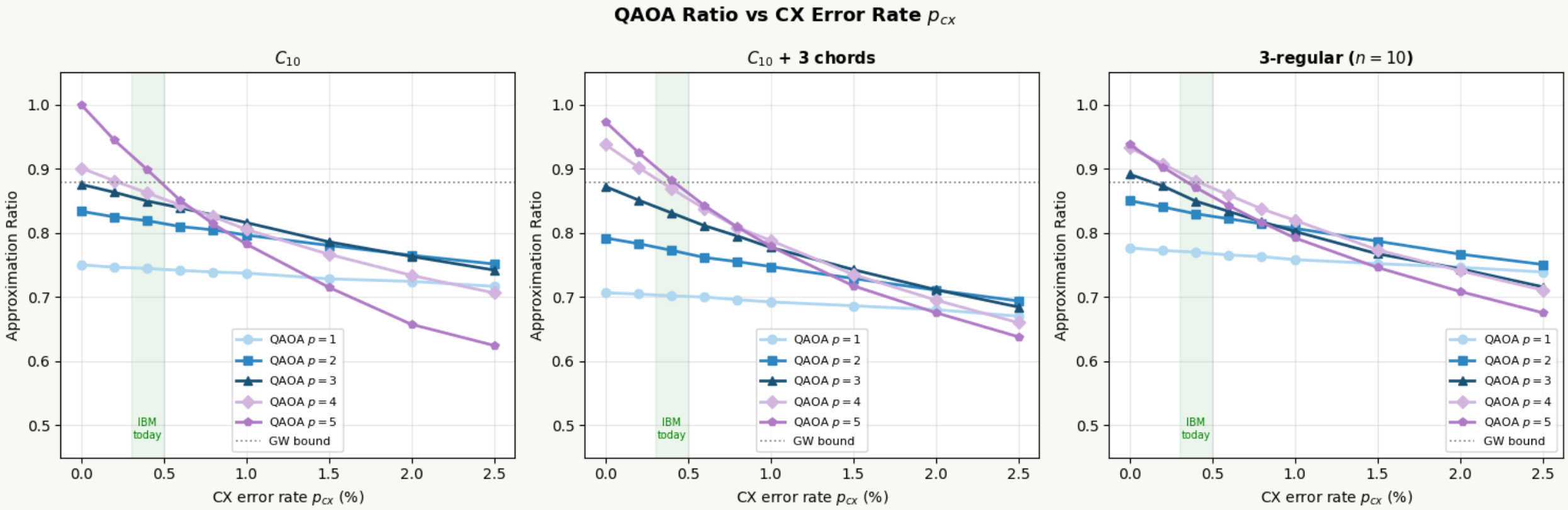
Introducing noise

Noise model

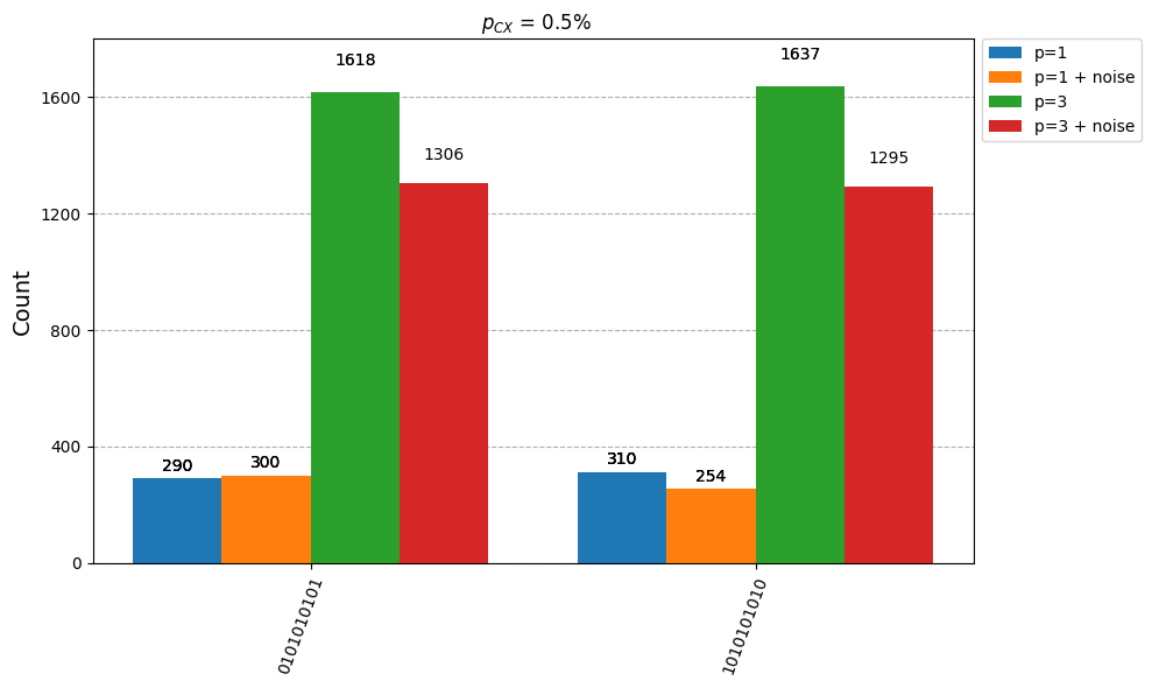
After each CX gate, apply the 2-qubit depolarizing channel with error probability p_{CX}

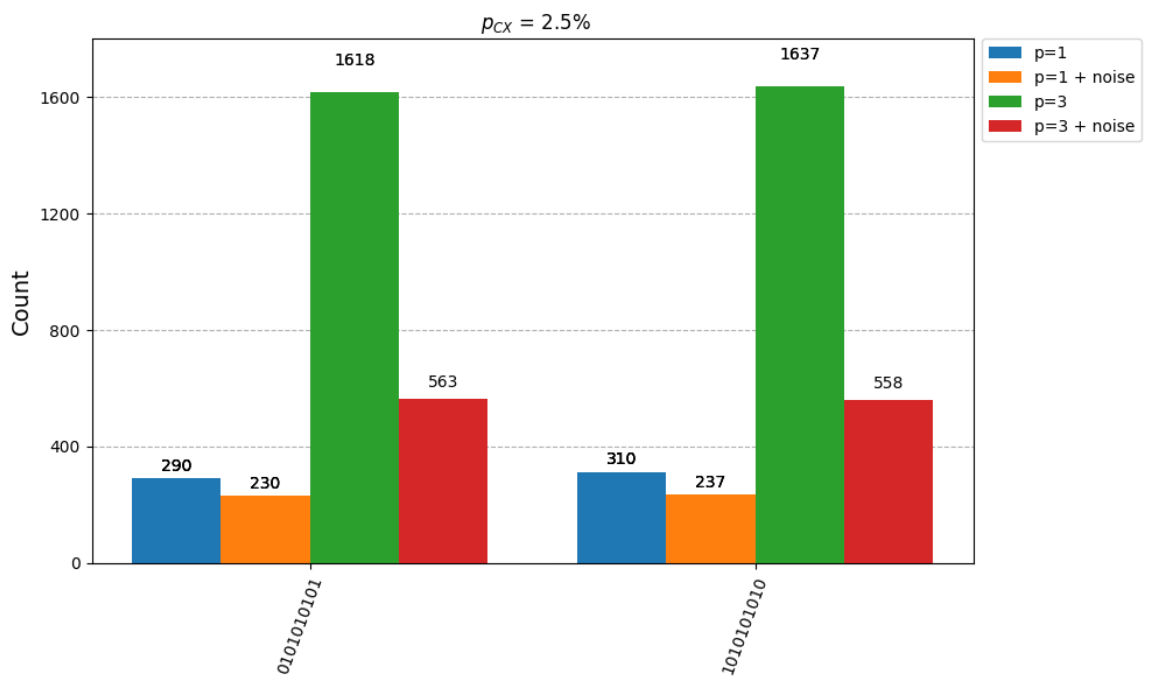
$$\varepsilon(\rho) = (1 - p_{CX})\rho + \frac{p_{CX}}{15} \sum_{P \epsilon \{I,X,Y,Z\}^{\otimes 2} \setminus \{I^{\otimes 2}\}} P\rho P^{\dagger}$$

Ratio evolution with noise



Noise in C_{10}


p_{CX} = 0.5%
Count
0
400
800
1200
1600
290
300
1618
1306
310
254
1637
1295
0101010101
1010101010
p=1
p=1 + noise
p=3
p=3 + noise


p_{CX} = 2.5%
Count
0
400
800
1200
1600
290
230
1618
563
310
237
1637
558
0101010101
1010101010
p=1
p=1 + noise
p=3
p=3 + noise

QAOA for MaxCut

- MaxCut as a way of finding 2 groups with as much correlation as possible.
- QAOA can perform this function at least as good as classical computers.
- Classical-Quantum hybrid algorithm.
- Circuit and results for n=10 qubits with different distributions (edges).
- Better performance for $p \rightarrow \infty$.
- Limits of performance based on noise.

We haven't proof quantum advantage yet, but we haven't proof it is not possible

QUANTUM APPROXIMATE OPTIMIZATION ALGORITHM

Thanks for your attention!

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